INCH POUND
MIL-DTL-2953D
6 June 2018
SUPERSEDING
MIL-S-2953C
4 February 1980

DETAIL SPECIFICATION

STRAINERS, STEAM (SIZES 3 INCHES AND BELOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers basket-type steam strainers in sizes 3 inches and below.

1.2 Classification. Strainers covered by this specification are of the compositions, ratings, end styles, and sizes specified (see 1.2.1 through 1.2.4 and 6.2).

1.2.1 Composition code letter. The material of the strainer is identified as follows (see 3.1):

Composition	Composition code letter
Composition A	A
Composition B	B
Composition D	D

1.2.2 Rating code number. The pressure rating of the strainer is identified as follows:

ASME B16.5 rating	Rating code number
Class 600 pound	06
Class 1500 pound	15

1.2.3 End style code letter. The type of strainer end connections is identified as follows (see 3.2.5):

End connection type	End style code letter
Flanged	F
Butt-Weld	B
Socket-Weld	S

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1.2.4 Size code number. The strainer size is identified as follows (see 3.2.1):

Nominal pipe size	Size code number
½	08
¾	12
1	16
1¼	20
1½	24
2	32
2½	40
3	48

1.3 Part or identifying number (PIN). PINs to be used for strainers acquired to this specification are created as follows:

M	**2953**	**-**	**X**	**XX**	**X**	**XX**
Prefix for Military Specification	Specification Number		Composition code letter (see 1.2.1)	Rating code number (see 1.2.2)	End style code number (see 1.2.3)	Size code number (see 1.2.4)

Example: An example PIN created for a ½-inch, composition A, 600-pound rated, flanged strainer is M2953-A06F08.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for

MIL-G-24716 - Gaskets, Metallic-Flexible Graphite, Spiral Wound

(Copies of these documents are available online at http://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A193/A193M - Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service

ASTM A194/A194M - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High-Temperature Service, or Both

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Welding, for High-Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martenistic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A696 - Standard Specification for Steel Bars, Carbon, Hot-Wrought or Cold-Finished, Special Quality, for Pressure Piping Components

(Copies of these documents are available online at www.astm.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.12 - Class 5 Interference – Fit Thread

ASME B16.5 - Steel Pipe Flanges and Fittings

ASME B16.11 - Forged Steel Fittings, Socket-Welding and Threaded

ASME B16.25 - Buttwelding Ends

ASME Boiler and Pressure Vessel Code Section VIII

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified in table I.

TABLE I. Materials.

Name of part	Materials form	Composition A	Composition B	Composition D
Body	Casting	ASTM A217/A217M, grade WC9	ASTM A217/A217M, grade WC6	ASTM A216/A216M, grade WCB
Cover	Casting	ASTM A217/A217M, grade WC9	ASTM A217/A217M, grade WC9	ASTM A217/A217M, grade WC9
	Bar stock	------	------	ASTM A696
Basket screen	------	ASTM A240/A240M	ASTM A240/A240M	ASTM A240/A240M
Gaskets	Spiral wound	MIL-G-24716, type I	MIL-G-24716, type I	MIL-G-24716, type I
Studs for covers	Alloy steel for high temperature bolting	ASTM A193/A193M, grade B16	ASTM A193/A193M, grade B16	ASTM A193/A193M, grade B16
Nuts for covers	Heavy semi-finished hexagon carbon and alloy steel for high temperature bolting	ASTM A194/A194M, grade 4	ASTM A194/A194M, grade 4	ASTM A194/A194M, grades 2, 2H, and 4

3.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Design and construction.

3.2.1 Sizes. Strainers shall be furnished in nominal pipe sizes of ½, ¾, 1, 1¼, 1½, 2, 2½, or 3, as specified (see 1.2 and 6.2).

3.2.2 End connections. Strainers shall have flanged ends in accordance with ASME B16.5 and table I, butt-weld ends in accordance with ASME B16.25, or socket weld ends in accordance with ASME B16.11, as specified (see 1.2 and 6.2). The design of the strainer nozzle cross-section, and the area between the end connection and the strainer body, shall be such that the connecting piping (considering maximum size and schedule to be used) will reach yield stress prior to the strainer nozzle cross-section. Face-to-face dimensions for flanges shall be in accordance with table II.

TABLE II. Flange face-to-face dimensions.

ASME B16.5 rating	Strainer size (inches)							
	½	¾	1	1¼	1½	2	2½	3
Class 600	5⅝	6⅜	7⅜	6⅜	9⅞	11⅞	13⅜	14⅜
Class 1500	11	11¼	11½	13	14¼	17¼	20	23

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3.2.3 Covers. Covers shall be flanged and bolted. Covers shall not be fitted with any connection for a blow valve. Wye type strainer covers shall be provided with a boss to facilitate attachment of a blow down or drain connection. The following features shall be incorporated:

a. Cover alignment shall be obtained by body guiding, that is, a close tolerance fit between the cover and strainer body. The cover shall not be aligned by means of the cover fasteners.

b. Flange joint design shall assure proper compression of type I gaskets, in accordance with MIL-G-24716, when body and cover flanges are drawn up metal-to-metal.

c. Cover flanges shall be secured by any of the following:

(1) Through bolts threaded the entire length and fitted with a nut on each end. Threads on the bolts and nuts shall be class 2 fit.

(2) Tap end studs with interference class 5 fit at tap end and a class 2½ fit at nut end. Interference shall be in accordance with ASME B1.12.

(3) Cap screws in accordance with B-7, class 2 fit of ASTM A193/A193M. Allen head type cap screws shall not be used.

d. Threaded fasteners, quantity and size, shall comply with ASME Boiler and Pressure Vessel Code, Section VIII, Rules for Bolted Flange Connections with Ring Type Gaskets.

3.2.4 Basket screen. The free area through the basket screen shall be at least 2 times the area of the inlet. Holes shall be either punched or drilled with all burrs removed. Holes shall be 0.020 inch in diameter and equally spaced over the basket. Spacing of holes on centers shall be approximately twice the diameter of the hole or greater. Metal thickness of basket shall be 0.015 inch minimum. Wire mesh shall not be used.

3.2.5 Strainer body. The design of all strainers shall provide that steam shall pass through the basket and shall be directed axially into the inside of the basket so that any foreign material intercepted by the strainer may be withdrawn by removal of the cover. The basket shall be installed in such a manner as to provide for easy removal. Each strainer shall have distinctly cast or stamped on one side of the body the size, the trademark of the manufacturer, and the pressure rating of the strainer. The walls of the bodies shall be curved surfaces; no flat surfaces shall be permitted. They shall be ribbed as necessary to provide strength. The low point of the strainer body shall be provided with a boss to facilitate installation of a drain connection by welding. In wye type strainers the boss on the strainer cover is considered to fulfill this requirement.

3.3 Pressure temperature. Pressure temperature rating shall be in accordance with ASME B16.5, class 600 and class 1500.

3.4 Shock. Strainers furnished in accordance with this specification shall meet the requirements of MIL-DTL-901 when tested in accordance with 4.3.2.

3.5 Nondestructive testing. Strainers shall be 100 percent radiographically inspected in accordance with S9074-AR-GIB-010/278.

4. VERIFICATION

4.1 Sampling for examination. A random sample of strainers shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.1.1 Visual examination. Sample strainers shall be visually examined in accordance with table III. Strainers not meeting the requirements of table III shall be rejected.

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TABLE III. Visual examination requirements.

Examination	Test paragraph	Requirement paragraph
Dimensions	4.1.2	3.2.1, 3.2.2
Materials	4.1.3	3.1
Design and construction	4.1.3	3.2
Pressure and temperature	4.1.3	3.3

4.1.2 Dimensional examination. Sample strainers shall be examined to determine conformance with the dimensional requirements specified herein (see 3.2.1 and 3.2.2).

4.1.3 Design. Sample strainers shall be examined to determine conformance with the material (see 3.1), design and construction (see 3.2), and pressure and temperature (see 3.3) requirements specified herein.

4.3 Tests.

4.3.1 Shell tests. Each of the strainers shall be subjected to the pressures specified in table IV for the duration of 1 minute. The water temperature shall not exceed 100 °F. Any weeping, leakage, or permanent deformation shall be cause for rejection.

TABLE IV. Test pressures for strength and porosity.

ASME B16.5 rating	Pressure (lb/in^2)	
	Composition A and B	Composition D
Class 600	2,250	2,225
Class 1500	5,625	5,575

4.3.2 Shock. Equipment and test classification shall be grade A, hull/bulkhead mounted, class III, type A. Method of mounting shall comply with figure 1.

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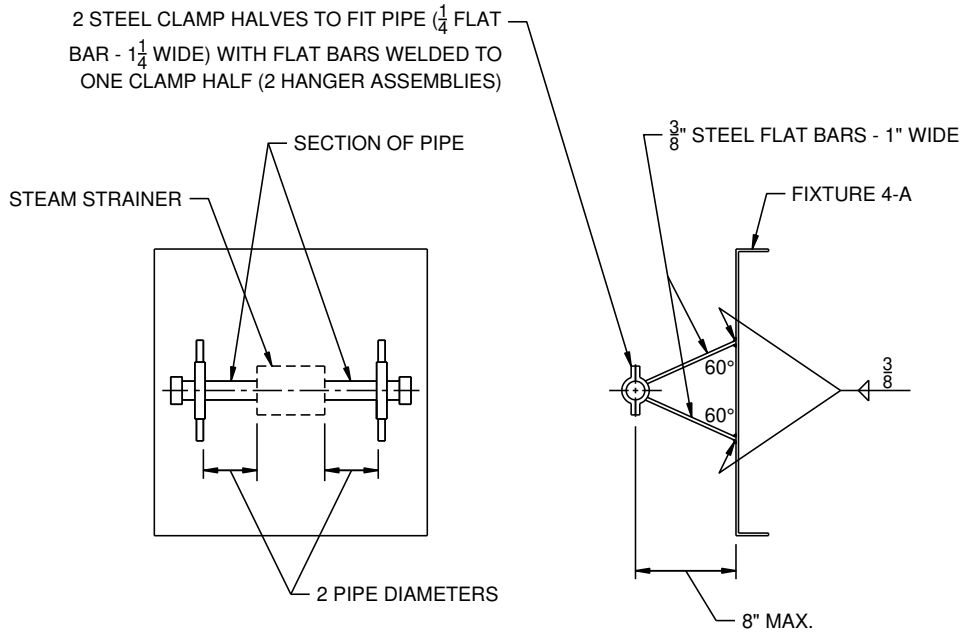


FIGURE 1. Straight tube, floating head, removable bundle, outside packed stuffing box.

4.3.2.1 Hydrostatic Test. Strainer shall be hydrostatically pressurized with water to the cold rating specified in ASME B16.5. Any external leakage of the strainer or strainer/piping or visible damage to the strainer basket shall constitute failure.

4.3.3 Nondestructive tests. Strainers shall be 100 percent radiographically inspected in accordance with S9074-AR-GIB-010/278.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. Strainers covered by this specification are intended for steam and steam drain use only and have pressure-temperature ratings in accordance with ASME B16.5 except for maximum allowable temperatures. Strainers need to have service lives that are commensurate with the service life of a naval ship, and will be used in an environment that is subject to shock loadings associated with military service. The maximum temperature limitations are 1,050 °F for composition A strainers, 1,000 °F for composition B strainers, and 775 °F for composition D strainers. Current stock of strainers may be used until depleted; future requisitions for strainers should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. PIN (see 1.3).

c. Design and construction requirements (see 3.2.1 and 3.2.2).

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d. Quantity required (3.2.3).

e. AQL of 1.0 (see 4.1).

f. Packaging requirements (see 5.1).

6.3 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired for incorporation in the equipment and lose their separate identity when the equipment is shipped.

6.4 Subject term (key word) listing.

Basket

Body

Cover

Pressure

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH
DLA – CC

Review activities:
Army – AR
Navy – YD

Preparing activity:
Navy – SH
(Project 4730-2018-066)

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